



Customer Shopping Behavior Analysis

Uncovering insights from **3,900 transactions** across product categories to guide strategic business decisions on spending patterns, customer segments, and subscription behavior.

Dataset at a Glance

3,900

Transactions

Rows in the dataset

18

Features

Columns per record

37

Missing Values

In Review Rating only

\$59.76

Avg Purchase

Mean transaction value

Key features span customer demographics (Age, Gender, Location, Subscription Status), purchase details (Item, Category, Amount, Season), and shopping behavior (Discount, Frequency, Review Rating, Shipping Type).

Data Preparation in Python

01

Load & Explore

Imported dataset via pandas; used `df.info()` and `.describe()` for structure and summary stats.

02

Clean Missing Data

Imputed 37 missing Review Rating values using the median rating per product category.

03

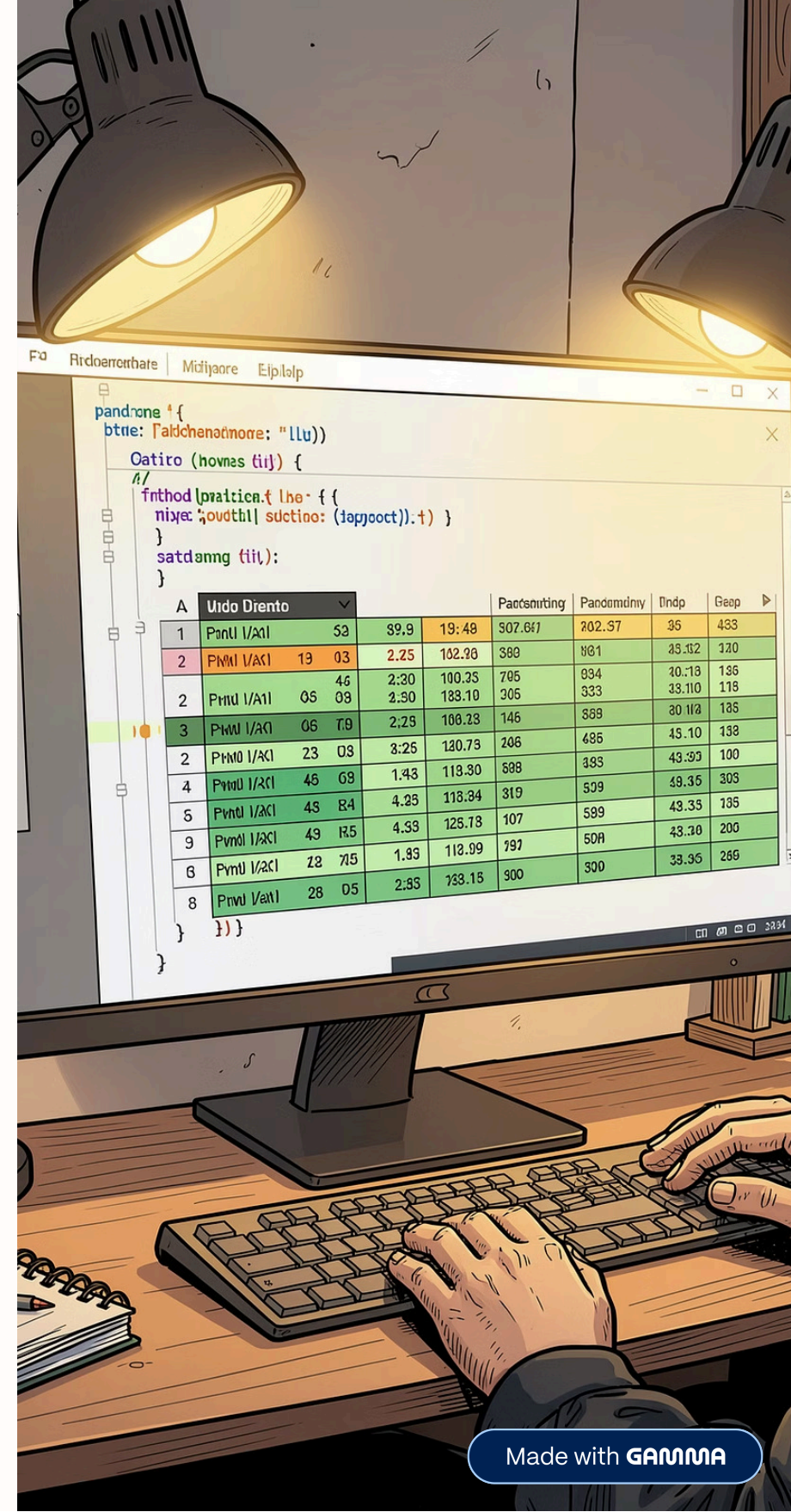
Feature Engineering

Created `age_group` bins and `purchase_frequency_days`; dropped redundant `promo_code_used`.

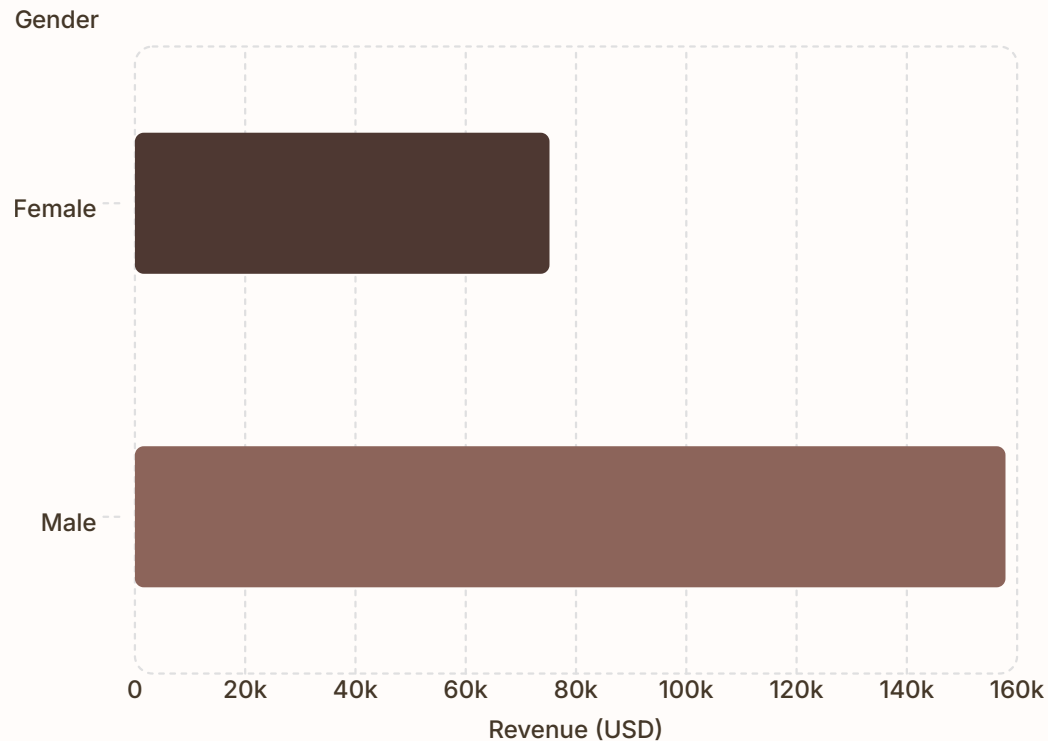
04

Database Integration

Loaded cleaned DataFrame into PostgreSQL for SQL analysis.



Revenue by Gender & Discount Users



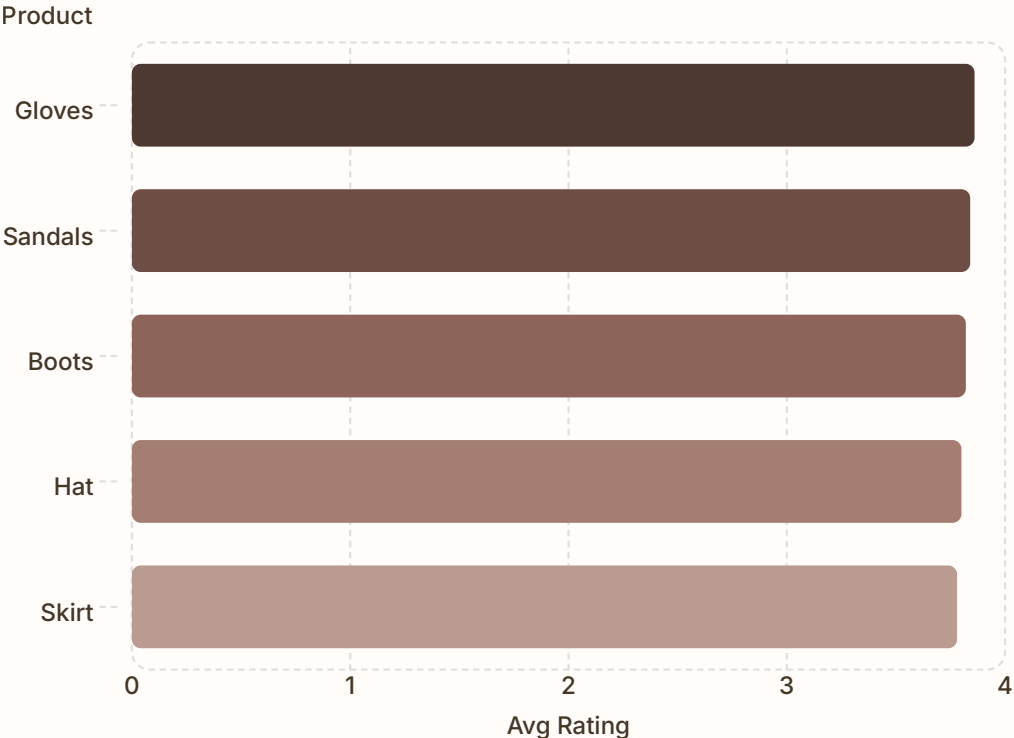
Key Findings

Male customers generated **more than double** the revenue of female customers — \$157,890 vs. \$75,191.

839 customers used discounts yet still spent above the average purchase amount, indicating strong price sensitivity paired with high intent.

Top Products & Shipping Insights

Top 5 Products by Average Rating



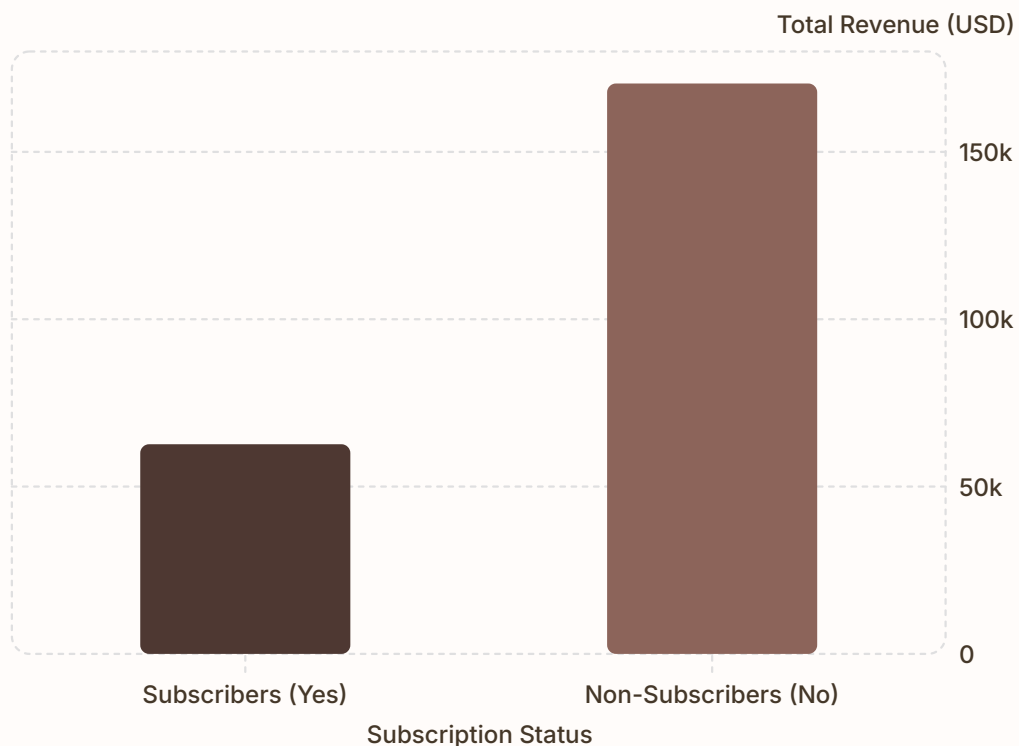
Shipping Type Comparison

Express shipping customers spend slightly more on average: **\$60.48** vs. **\$58.46** for Standard shipping.

Most Discount-Dependent Products

- Hat — 50.00% discount rate
- Sneakers — 49.66%
- Coat — 49.07%
- Sweater — 48.17%
- Pants — 47.37%

Subscribers vs. Non-Subscribers



What the Numbers Reveal

Non-subscribers account for **73% of customers** and **\$170,436 in revenue**, despite nearly identical average spend (\$59.87 vs. \$59.49).

Only **27% of customers** are subscribed — a significant untapped opportunity for recurring revenue growth.

- ❏ Repeat buyers: 2,518 non-subscribers vs. 958 subscribers — loyalty does not automatically convert to subscriptions.

Customer Segmentation

Loyal Customers

3,116 customers — the dominant segment, representing the core repeat-purchase base.

Returning Customers

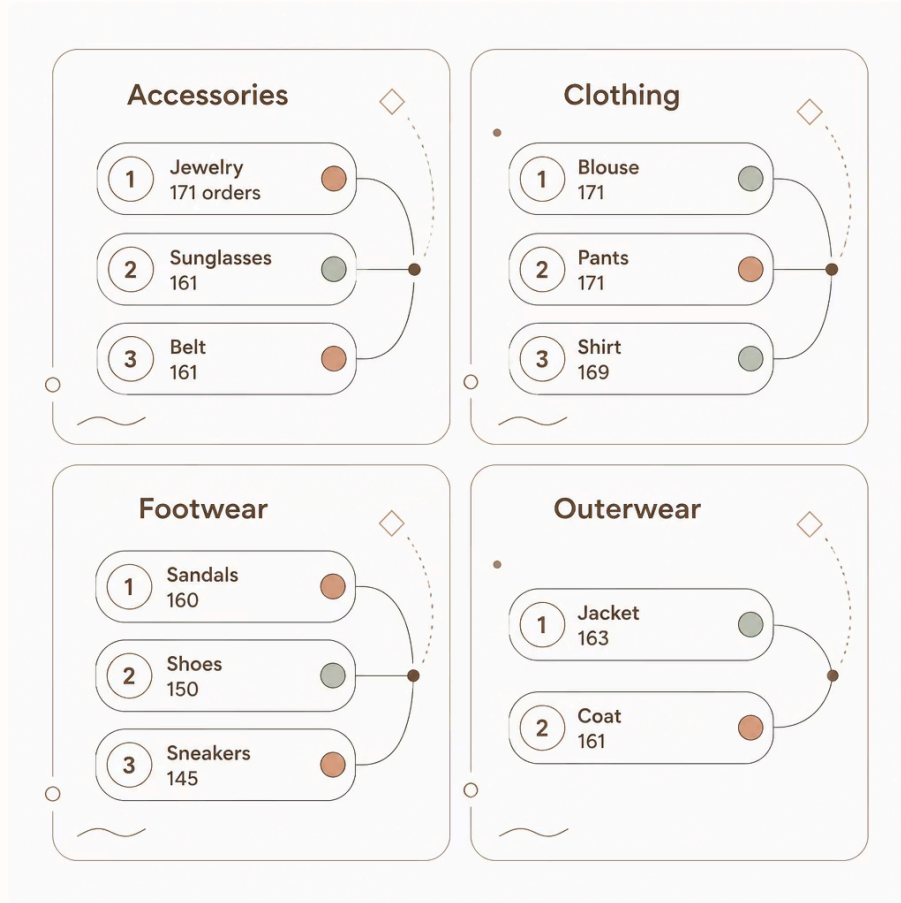
701 customers — engaged but not yet fully loyal; prime candidates for retention programs.

New Customers

83 customers — smallest segment; conversion to returning buyers is the key growth lever.



Top Products by Category

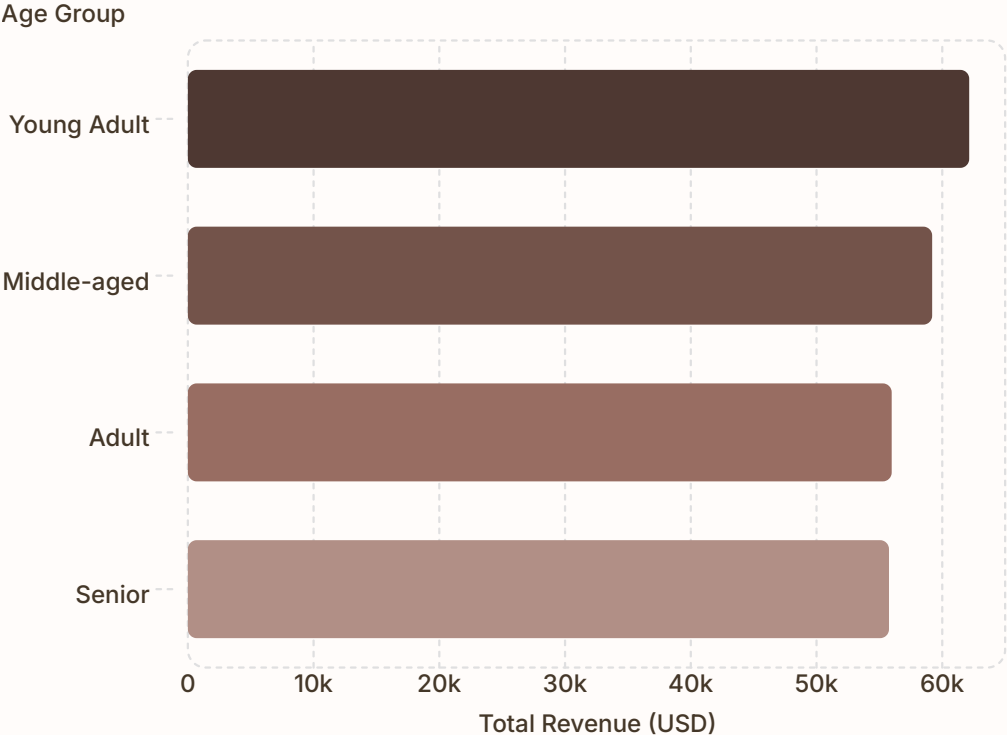


Category Leaders

Clothing leads with **Blouse, Pants, and Shirt** each at ~170 orders. In Outerwear, **Jacket** (163) and **Coat** (161) are top sellers. Footwear is led by **Sandals** (160), and Accessories by **Jewelry** (171).

These top performers should be prioritized in marketing campaigns and inventory planning.

Revenue by Age Group



Age Group Insights

Young Adults lead in revenue contribution at \$62,143, followed closely by Middle-aged (\$59,197). Revenue is relatively evenly distributed across all four age groups, suggesting broad market appeal.

Targeted marketing toward Young Adults and express-shipping users offers the highest ROI potential.



Dashboard & Business Recommendations



Power BI Dashboard

Interactive dashboard tracking 3.9K customers, \$59.76 avg purchase, 3.75 avg rating — with filters for subscription, gender, category, and shipping.



Boost Subscriptions

Promote exclusive subscriber benefits to convert the 73% non-subscriber base.



Loyalty Programs

Reward repeat buyers to move them into the "Loyal" segment (currently 3,116 customers).



Targeted Marketing

Focus on high-revenue age groups, express-shipping users, and top-rated products in campaigns.